

#	Function	Sub-function	Description	Architectural Significant Requirements
1	Authentication & Access	Sign Up	Người dùng đăng ký tài khoản mới	- Security: Enforce strong password policy, secure password hashing, HTTPS/TLS, email verification, rate limiting. - Performance: Registration request must complete within 5 seconds. - Reliability: Prevent duplicate registrations and support account verification flow. - Auditability: Log registration attempts, IP address, timestamp, and result.
2	Authentication & Access	Sign In	Người dùng đăng nhập hệ thống	- Security: Secure authentication, brute-force protection, session expiration, refresh token rotation. - Performance: Login response time < 2 seconds. - Reliability: Failed login protection and secure session management. - Auditability: Log login attempts, account lock events, and IP metadata.
3	Authentication & Access	Sign Out	Người dùng đăng xuất khỏi hệ thống	- Security: Session invalidation and secure token revocation. - Performance: Logout operation < 1 second. - Auditability: Log logout event and session identifier.
4	Authentication & Access	OAuth2 Login	Đăng nhập bằng tài khoản Google	- Security: OAuth2/OpenID Connect support, CSRF protection, token validation. - Interoperability: Support third-party identity providers. - Auditability: Log provider, email, and login metadata.
5	Authentication & Access	OTP Verification	Xác thực OTP qua email	Security: OTP expiration, single-use validation, rate limiting. Performance: OTP delivery within 30–60 seconds. Auditability: Log OTP generation and verification events.
6	Authentication & Access	Forgot / Reset Password	Đặt lại mật khẩu	Security: Secure reset tokens, HTTPS transmission, strong password policy. Reliability: Invalidate active sessions after password reset. Auditability: Log reset requests and completion status.
7	User Profile	View / Update Profile	Người dùng quản lý hồ sơ cá nhân	Security: Users may only access and update their own profile. Performance: Profile operations < 2 seconds. Auditability: Log profile changes and session activities.
8	User Profile	Upload Avatar	Tải ảnh đại diện	Security: Validate file type, file size, and upload integrity. Performance: Upload operation < 5 seconds. Interoperability: Support cloud storage and CDN integration.
9	User Profile	Manage Addresses	Quản lý địa chỉ giao hàng	Security: Ownership validation and secure data handling. Performance: CRUD operations < 2 seconds. Interoperability: Integrate external geographic/address services.
10	Product Catalog	Browse Products	Xem danh sách sản phẩm	Security: Public API protection and anti-scraping measures. Performance: Product listing < 2 seconds with pagination and caching. Scalability: Support distributed caching and high concurrency.
11	Product Catalog	View Product Detail	Xem chi tiết sản phẩm	Security: Product pricing and inventory validated server-side. Performance: Product detail page < 2 seconds. Interoperability: Aggregate data from multiple services/modules.
12	Search & Discovery	Search Products	Tìm kiếm sản phẩm	Security: Validate and sanitize search queries. Performance: Search response < 1.5 seconds. Scalability: Full-text indexing and distributed search support. Auditability: Log search analytics.
13	Search & Discovery	Autocomplete	Gợi ý từ khóa tìm kiếm	Performance: Suggestion response < 200 milliseconds. Scalability: Support caching and prefix indexing.
14	Recommendations	Personalized Recommendations	Gợi ý sản phẩm cá nhân hóa	Security: Recommendation access requires authenticated user context. Scalability: Support distributed recommendation engine and event ingestion. Performance: Cached recommendation responses.

15	AI Shopping Assistant	Chat with AI	AI hỗ trợ tư vấn sản phẩm	Security: Prompt sanitization, API key protection, prompt injection prevention. Performance: Streaming AI responses with timeout handling. Reliability: Graceful fallback during AI service failure. Auditability: Log AI interactions and token usage.
16	Shopping Cart	Manage Cart	Quản lý giỏ hàng	Security: Authenticated cart ownership validation. Performance: Cart operations < 1 second. Reliability: Support cart persistence and merge strategies. Auditability: Log cart modifications.
17	Checkout	Proceed to Checkout	Xác nhận đơn hàng	Security: Revalidate pricing and inventory during checkout. Performance: Checkout response < 2 seconds. Reliability: Transactional consistency for order creation.
18	Checkout	Apply Coupon / Voucher	Áp dụng mã giảm giá	Security: Server-side discount validation and anti-abuse controls. Performance: Validation < 1 second.
19	Checkout	Place Order	Tạo đơn hàng	Security: Secure order processing and PII protection. Reliability: Atomic transaction for order creation and inventory reservation. Performance: Order creation < 5 seconds. Auditability: Full order lifecycle logging.
20	Checkout	Pay via Payment Gateway	Thanh toán VNPAY / MoMo	Security: Secure payment callback validation and HTTPS communication. Reliability: Idempotent callback handling and retry support. Interoperability: Payment gateway integration support. Auditability: Log payment transactions and gateway responses.
21	Order Management	View Orders	Xem danh sách đơn hàng	Security: Users may only access their own orders. Performance: Order detail < 1 second with pagination.
22	Order Management	Cancel Order	Hủy đơn hàng	Security: Validate order ownership and order state. Reliability: Atomic cancellation and inventory restoration. Auditability: Log cancellation reason and status changes.
23	Refunds & Disputes	Request Refund	Yêu cầu hoàn tiền	Security: Refund ownership validation and state validation. Reliability: Synchronize refund status with payment services. Auditability: Full refund lifecycle logging.
24	Reviews & Ratings	Write / Edit Review	Đánh giá sản phẩm	Security: Only verified purchasers may submit reviews. Reliability: Prevent duplicate reviews. Performance: Review submission < 2 seconds. Auditability: Log review activities.
25	Chat & Notifications	Send Message	Nhắn tin với seller/support	Security: Conversation ownership validation and message sanitization. Reliability: Reliable message delivery. Performance: Message delivery < 1 second. Auditability: Log communication metadata.
26	Seller Platform	Seller Registration	Đăng ký seller	Security: Validate seller documents and protect sensitive information. Performance: Seller registration < 5 seconds. Auditability: Log seller onboarding activities.
27	Seller Product Management	Create / Edit Product	Seller quản lý sản phẩm	Security: RBAC authorization and input validation. Reliability: Atomic save operations for products and SKUs. Performance: Product save < 10 seconds. Auditability: Log product modifications.
28	Seller Inventory	Adjust Inventory	Điều chỉnh tồn kho	Security: Validate ownership and inventory integrity. Reliability: Prevent race conditions and support transactional inventory updates. Auditability: Log inventory adjustments and stock changes.
29	Seller Finance	Request Payout	Yêu cầu rút tiền	Security: Validate seller identity and payout eligibility. Reliability: Idempotent payout requests and transactional consistency. Auditability: Log payout requests and financial changes.

30	Admin Dashboard	View Analytics	Dashboard thống kê	Security: RBAC enforcement for admin access. Performance: Cached analytics and asynchronous aggregation. Scalability: Support read replicas and large-scale analytics.
31	Admin User Management	Manage Users	Quản lý người dùng	Security: Strict RBAC and least-privilege access control. Performance: Indexed search operations. Auditability: Log all administrative actions on user accounts.
32	Admin Seller Management	Approve / Reject Seller	Quản lý seller	Security: State validation and administrative authorization. Reliability: Atomic seller approval workflows. Auditability: Log seller moderation decisions.
33	Admin Product Moderation	Approve / Reject Product	Kiểm duyệt sản phẩm	Security: RBAC authorization and moderation controls. Reliability: Atomic moderation operations. Auditability: Log moderation decisions and reasons.
34	Admin Refund Management	Process Refund	Xử lý hoàn tiền	Security: Administrative authorization and approval workflows. Reliability: Atomic refund processing and payment synchronization. Auditability: Full refund processing traceability.
35	Admin Audit	View Audit Logs	Xem nhật ký hệ thống	Security: Immutable audit logs and restricted access. Scalability: Long-term log retention and full-text search support. Compliance: Support regulatory compliance requirements.
36	Cross-Cutting Architecture	Security Architecture	Kiến trúc bảo mật toàn hệ thống	Security: RBAC, HTTPS/TLS, encryption at rest, audit logging, API rate limiting.
37	Cross-Cutting Architecture	Performance Architecture	Kiến trúc hiệu năng	Performance: Horizontal scalability, distributed caching, pagination, read replicas.
38	Cross-Cutting Architecture	Reliability Architecture	Kiến trúc độ tin cậy	Reliability: Transactional consistency, retry strategies, rollback handling, graceful degradation.
39	Cross-Cutting Architecture	Observability & Monitoring	Logging & Monitoring	Auditability: Centralized logging, metrics collection, monitoring, alerting, searchable logs.
40	Cross-Cutting Architecture	Compliance Requirements	Tuân thủ quy định	Compliance: Personal data protection, payment compliance, long-term audit retention.
				\